Identification and Control of Motor-Rotor and Motor-Pendulum Systems

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1 Introduction

Modeling of physical systems is one of the most essential aspects of designing control algorithms for implementation. Collecting data about the physical system is crucial for generating a good approximation of the actual system. The rate at which the state of the system to be controlled is estimated plays an important role in generating a good model. Further, systems may be subjected to disturbances from a variety of factors, which need to be accounted for by the control design. Additionally, the continuous-time nature of physical system and discrete-time nature of devices used to control them present a new gap that needs to be bridged.

In this study, we consider three physical systems - a rotor, a stable pendulum, and an unstable inverse pendulum. System identification is done using an aluminum bar attached at its center to a DC motor and is further extended to more complex models such as a stable pendulum and inverted pendulum. A state observer feedback control is implemented in the system.

2 Motor-Rotor System

The plant for a motor-rotor system approximated by a simple first-order system is shown in Figure 1

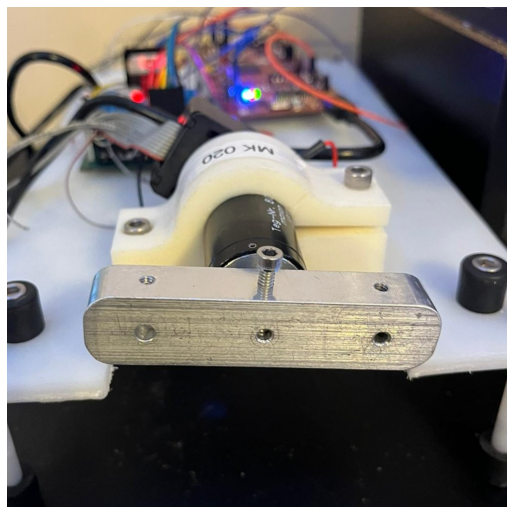

Figure 1: Plant under analysis

The system contains a DC motor attached to a solid aluminum block with 3 through holes for axle and 3 perpendicular threaded holes for set screws. Texas Instruments C2000 microcontroller is used in the system along with a motor driver to control the motor. The control system is designed in Simulink R2022b and implemented on the system.

2.1 Motor-Rotor Model

A CAD model of the aluminum rod attached is shown in Figure 2
The governing equations of the electromechanical system are given below:

$$\frac{d\theta}{dt} = \omega \tag{1}$$

$$J_{rotor}\frac{d\omega}{dt} = -b\omega + K_m i - T_d \tag{2}$$

$$L\frac{di}{dt} = -Ri - K_m\omega + V_s u, (-1 < u < 1) \tag{3}$$

$$J_{rotor} = J_{motor} + J_{rod} \tag{4}$$

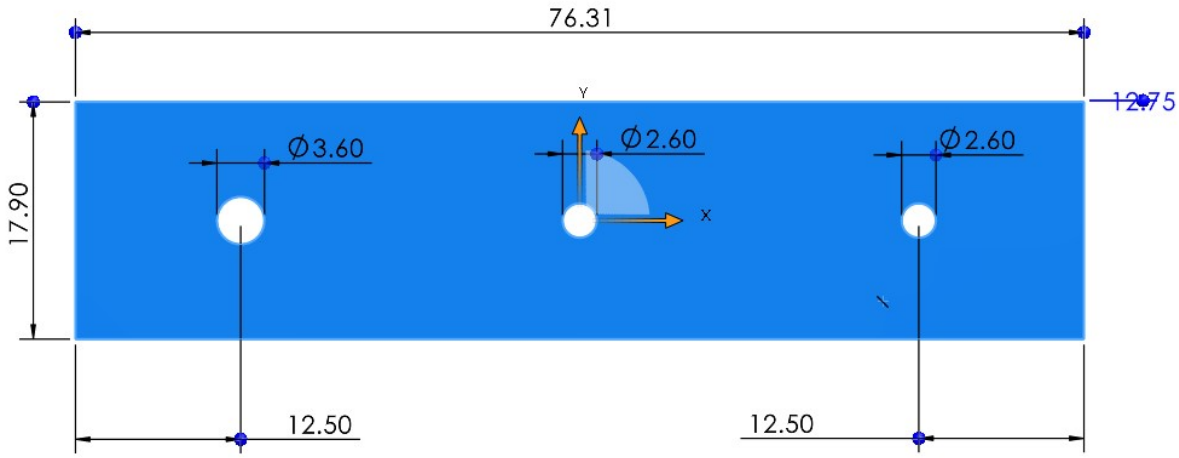

Figure 2: CAD model of aluminum rod

Here, the control input u is the duty cycle of the motor, and the disturbance torque $T_d = 0$. Taking the Laplace transform of the equations and solving for the transfer function from the duty cycle to the angular velocity gives the following result:

$$\frac{\omega(s)}{u(s)} = \frac{K_m V_s}{(J_{rotor}s + b)(Ls + R) + K_m^2} \tag{5}$$

To approximate it to a first-order system, the inductance, L is assumed to be negligibly small, i.e. $L \cong 0$. This assumption implies that the current through the system is constant (Equation 3), which is reasonable for a no-load system (this study), but it will fail for systems where external load is attached to the system, since the current will not regulate itself.
Equation 5 can be rewritten as

$$\frac{\omega(s)}{u(s)} \cong \frac{K_m V_s}{J_{rotor}Rs + bR + K_m^2} = \frac{\frac{K_m V_s}{bR+K_m^2}}{\frac{J_{rotor}Rs}{bR+K_m^2} + 1} = \frac{\kappa}{\tau s + 1} \tag{6}$$

2.2 System Identification

The system is identified using a "grey box" technique. A series of experiments were conducted on the motor-rotor system to identify parameters shown in equation 6. The methods of estimation of the various parameters are described below:

2.2.1 Mass

The aluminium rod was measured multiples times on a weighing scale, and the average was used as the mass:

$$m = 0.032kg \tag{7}$$

2.2.2 Moment of Inertia

The dimensions of the rod were measured using Vernier Calipers. Using the average values of the measurements, the following values were obtained:

$$l \times b \times h = 76.31 \times 17.90 \times 12.75mm^3$$

The moment of inertia is calculated analytically as:

$$J_{rod} = \frac{m(l^2 + b^2)}{12} = 16.383 * 10^{-6} kgm^2 \tag{8}$$

To verify, the moment of inertia was also estimated using a SolidWorks CAD model, which came out to be $J_{rod} = 16.393 \times 10^{-6} kgm^2$. For our experiments, we use $J_{rod} = 16.383 \times 10^{-6} kgm^2$.
The motor inertia is taken from the datasheet as $J_{motor} = 1.67 \times 10^{-6} kgm^2$.

$$J_{rotor} = J_{motor} + J_{rod} \tag{9}$$

$$J_{rotor} = 1.8053 \times 10^{-5} kgm^2 \tag{10}$$

2.2.3 Supply Voltage

The motor was supplied with an input duty cycle of 1, and voltage was measured across the terminals of the motor, which came out to be:

$$V_s = 10.5V \tag{11}$$

The drop in voltage is attributed to bias voltage of the 2 diodes in the motor driver. This is accounted for in the control programs using a 1D lookup table.

2.2.4 Motor Parameters

To obtain a response consistent with the transfer function, a duty cycle of 1 was commanded to the motor-rotor system twice, and rotor velocity was recorded using the encoder counts and plotted as shown in Figure 3

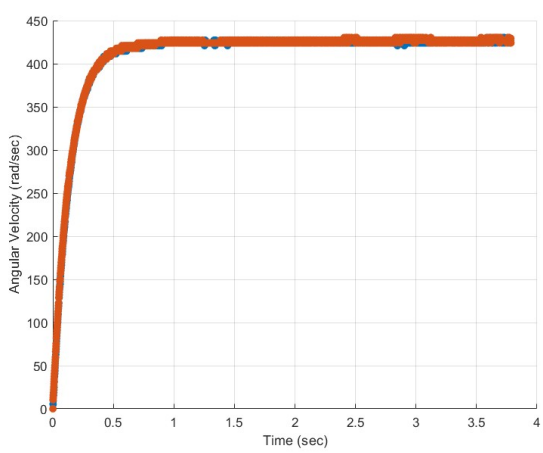

Figure 3: Experiment to determine plant model

A least-squares fit to the data collected gave the following parameters:

	Data 1	Data 2	Average
κ	427.1	426.9	427
τ	0.1399	0.1374	0.1387

Table 1: Estimation of Motor Parameters

To estimate the viscous damping coefficient b, an input duty of 100 percent (12V) was provided till the motor attained constant angular velocity, and then the input duty was switched to 0 percent, and the motor was

allowed to come to a complete stop. Considering equation 2, we observe that $T_d = 0$, and $i = 0$ since there is no input current while the motor is coming to a halt. Therefore, we have the equation

$$J_{rotor}\frac{d\omega}{dt} = -b\omega \tag{12}$$

$$\frac{dw}{\omega} = \frac{-b}{J_{rotor}}dt \tag{13}$$

$$\int \frac{dw}{\omega} = \int \frac{-b}{J_{rotor}}dt \tag{14}$$

$$\omega = \omega_{t_0} e^{\frac{-bt}{J_{rotor}}} \tag{15}$$

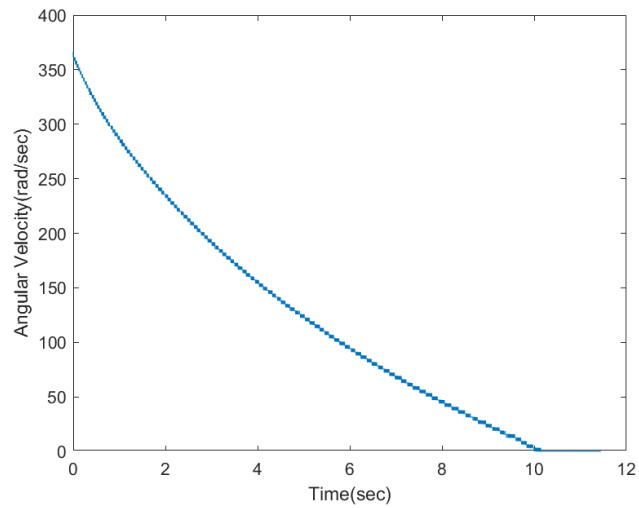

Figure 4: Experiment to determine damping coefficient

An exponential drop was predicted and experimental data verified the same. A first order exponentially decaying model $x_0 e^{\frac{-t}{\tau_{decay}}}$ was fit, and the time constant was found to be:

$$\tau_{decay} = 4.08$$

Comparing with the 15, and using 4, we get damping coefficient as

$$b = 4.425 \times 10^{-6}\frac{kgm^2}{s} \tag{16}$$

A multimeter was used to measure the resistance across the leads of the motor, and the average of the 3 readings was taken.

$$R = 6\Omega$$

Using data from Table 1 and comparing coefficients with equation 6, we get following results:

$$K_m = 2.5463 \times 10^{-2}$$

The parameters are summarised in the Table 2:

3 Extension of rotor model

The rotor configuration is extended to a pendulum configuration by attaching the motor axle to the right or left hole in the rod. The new system has a different moment of inertia, and disturbance torque T_d. However, the electromechanical equations 3 governing the motor behaviour remain the same.

Parameter	System ID	Measurement
Motor Constant K_m	2.459×10^{-2}	N/A
Supply Voltage V_s	N/A	10.5V
Terminal Resistance	N/A	6 Ω
Rotor Inertia J_{motor}	N/A	$1.8053 \times 10^{-5} kgm^2$
Inductance L	N/A	≈ 0
Viscous Damping b	$4.425 \times 10^{-6} \frac{kgm^2}{s}$	N/A
Encoder Resolution	N/A	400 counts/rev

Table 2: Parameters

3.1 Motor - Stable Pendulum Model

The stable pendulum system and a schematic of the system are shown below in Figure 5:

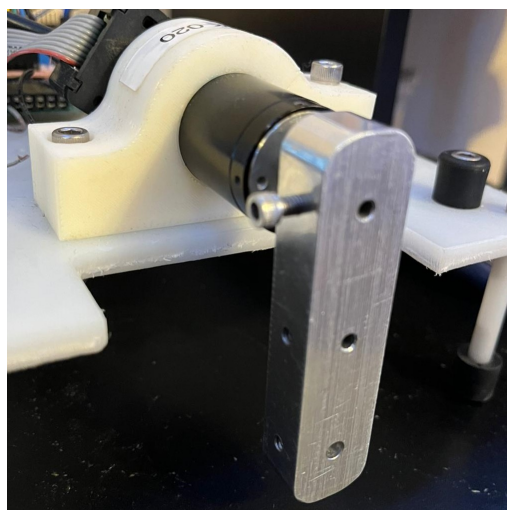

Figure 5: Stable Pendulum Configuration

Here, the physical parameters such as the moment of inertia and the disturbance torque will have values different from the motor-rotor case as shown below:

$$J_{pend} = J_{motor} + J_{rod} + ml_{dist}^2$$

$$T_d = -mgl_{dist} sin\theta$$

where $l_{dist} = 25.655mm$, and angle θ is measured from the vertical axis.
The following equations are obtained:

$$J_{pend}\frac{d\omega}{dt} = -b\omega + K_m i - mgl_{dist} sin\theta \tag{17}$$

$$L\frac{di}{dt} = -Ri - K_m\omega + V_s u \tag{18}$$

Using the small angle approximation, $sin\theta \cong \theta$, Laplace transforms were taken to get the transfer function from the input to angular velocity as:

$$\frac{\omega(s)}{u(s)} = \frac{K_m V_s}{(J_{pend}s^2 + bs + mgl_{dist})(Ls + R) + K_m^2} \cong \frac{K_m V_s}{JRs^2 + K_m^2 s + Rmgl_{dist}} = \frac{\kappa}{\tau s^2 + s + \frac{Rmgl_{dist}}{K_m^2}} \tag{19}$$

3.2 Motor - Unstable Pendulum Model

The unstable pendulum configuration is achieved by keeping the rod in a vertically upward position as shown in Figure 6

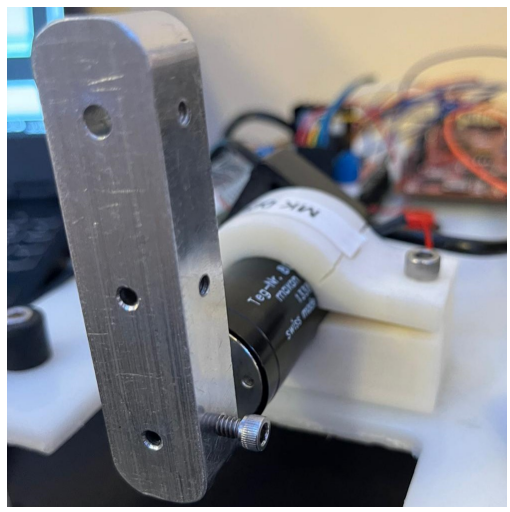

Figure 6: Unstable Pendulum Configuration

In the unstable pendulum case, the governing equations remain approximately the same as in the stable pendulum case - only the direction of the disturbance torque changes:

$$T_d = mgl_{dist}sin\theta$$

$$J_{pend}\frac{d\omega}{dt} = -b\omega + K_m i + mgl_{dist}sin\theta \tag{20}$$

$$L\frac{di}{dt} = -Ri - K_m\omega + V_s u \tag{21}$$

Using the small angle approximation, the transfer function from the input to the angular velocity is calculated as:

$$\frac{\omega(s)}{u(s)} = \frac{K_m V_s}{(J_{pend}s^2 + bs - mgl_{dist})(Ls + R) + K_m^2} \cong \frac{K_m V_s}{JRs^2 + K_m^2 s - Rmgl_{dist}} = \frac{\kappa}{\tau s^2 + s - \frac{Rmgl_{dist}}{K_m^2}} \tag{22}$$

3.3 System Model Validation using Chirp signal

The reduced order model of the motor-rotor system is a first-order system, so a step response was used to identify and develop a virtual plant model of the system. The linear and non-linear virtual plant model of motor-rotor system is then validated by studying the frequency response of the plant model and the system for a frequency sweep of $0.1 - 20hz$ using chirp-up signal. The linear, and non-linear models are found to be representative of the system from the frequency response of the system presented in the Fig. 7. From the Fig. 7, it can be seen that system has linear phase response at $0 - 20hz$ and is good operating range for the motor-rotor system for our application.
Using our knowledge of the system, the virtual plant model of the stable, and unstable pendulum are obtained changing the moment of inertia and linearizing the system as discussed in the earlier section. The modified

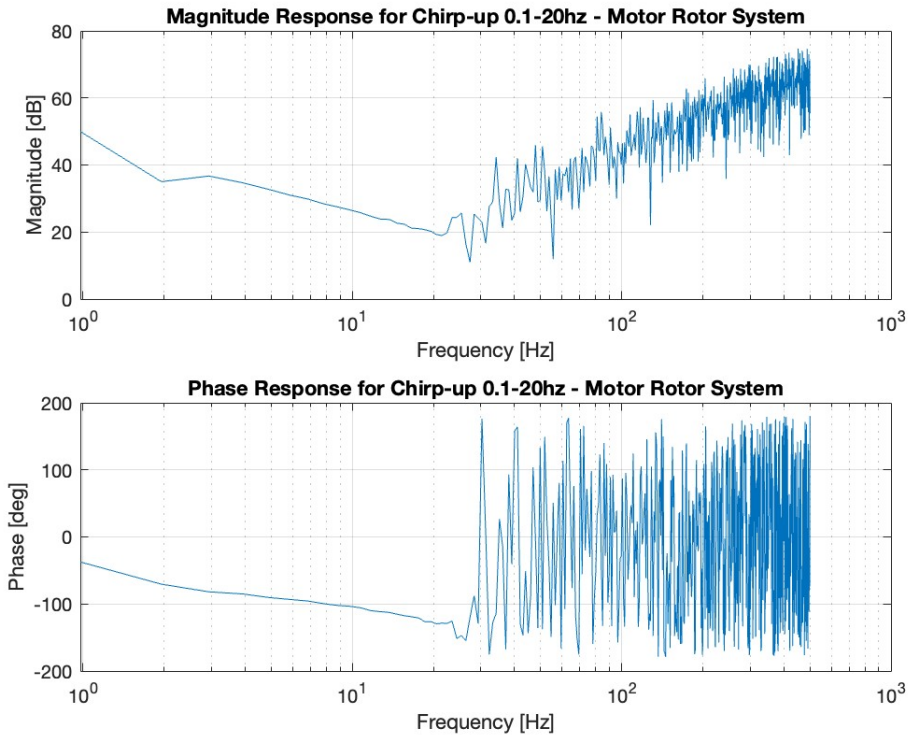

Figure 7: Chirp Response for Motor Rotor System

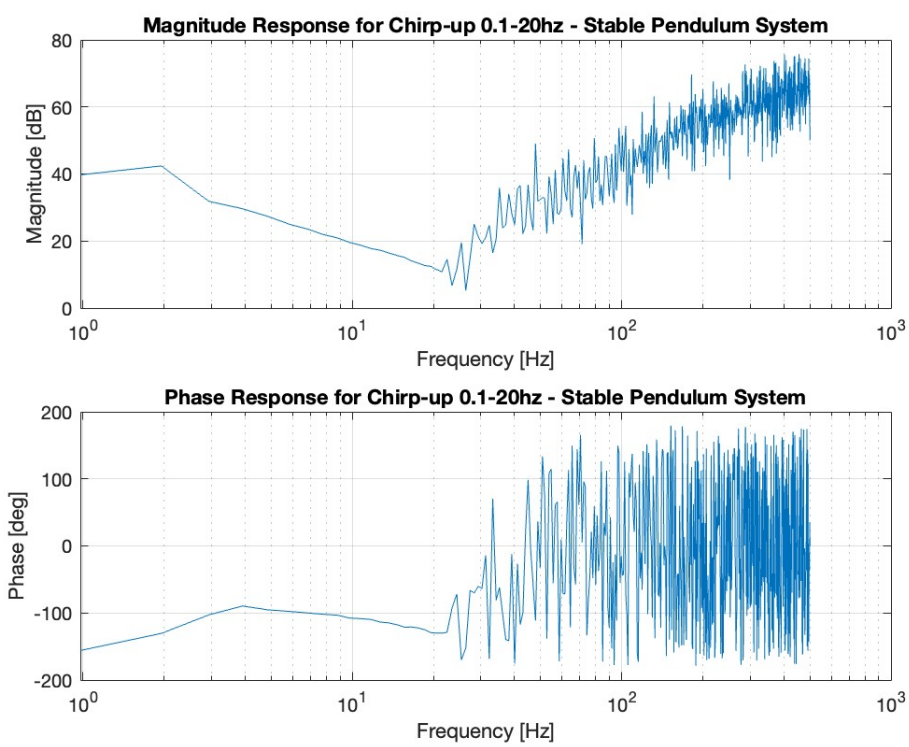

Figure 8: Chirp Response for Stable Pendulum Virtual Plant

Frequency	Observer Poles	State Feedback Poles
1000Hz	[0.5335, 0.5335]	[0.9811 + 0.0302i, 0.9811 - 0.0302i]
100Hz	[0.0019, 0.0019]	[0.7911 + 0.2518i, 0.7911 - 0.2518i]
10Hz	[5.16×10^{-28}, 5.16×10^{-28}]	[-0.1552 + 0.0093i, -0.1552 + 0.0093i]

Table 3: Poles of Closed Loop Systems

Frequency	Rise Time $t_r(s)$	Maximum Peak Overshoot $M_p(\%)$
1000Hz	0.0466	14.7
100Hz	0.0473	14.43
10Hz	0.134	8.42

Table 4: Step Response Characteristics for Closed Loop Motor Rotor System

plant model for stable pendulum and unstable pendulum differ by a sign and they are validated using chirp-up signal. The frequency response of the stable virtual plant is conformative with the stable pendulum plant as seen in Fig. 8. As in the case of motor-rotor system, the pendulum has desirable characteristics for the operating range $0 - 15Hz$.

4 Controller Design and Analysis

The position controller for the motor-rotor, stable pendulum and unstable pendulum systems are designed using State Observer Feedback Control (SOFC). Rise time t_r of $0.05s$ and maximum peak overshoot M_p of 15% is used to design the observer and feedback controller gains and they are presented in Table 3. The resulting closed loop systems are stable when sampling time T_s is $100hz$ and $10hz$ while a stable controller could not be designed for $T_s = 10hz$ for the required t_r and M_p. The poles of the resulting closed loop systems are presented in Table 3.

Table 4 showcases the step response characteristics for different closed loop configurations and are vital to understanding the performance of the systems. From the step response information, it can be seen that the performance of the $100hz$ and $1000hz$ systems meet our design specifications of $t_r = 0.05s$ and $M_p = 15\%$. However, a stable SOFC could not be designed for a $10hz$ sampled system to meet our desired specifications and this can be seen from the Table 4.

In order for the system to be stable the Nyquist plot should have at least one counter clockwise loop around the point -1 and this can be seen in the case of $100hz$ and $1000hz$ sampled motor rotor systems with SOFC as in Fig. 10, 9. Similarly, the $100hz$ and $1000hz$ sampled stable and unstable pendulum are also stable with SOFC. While the $10hz$ sampled systems satisfy the Nyquist criterion with SOFC as in Fig. 11, the response of real plant is unstable due to plant modelling errors and small vector gain margin as shown in Table 5. This is also evident in the simulation where the simplified linear model is stable and tracks the reference well while response of nonlinear model is unstable.

Although the closed loop systems are stable, they are very sensitive to process variations at the upper end of the operating range at $15 - 20hz$. This is seen from the sensitivity Bode plots in Fig. 12, 13, & 14 and is not desirable characteristic for practical applications.

Frequency	Open Loop VGM	Closed Loop VGM
1000Hz	7.2186	8.03
100Hz	4.9552	7.0131
10Hz	3.47	2.5417

Table 5: Vector Gain Margin for Sampled Motor Rotor System

The vector gain margins for the stable system configurations are presented in Table 5 are useful to understand the robustness of the closed loop system. As expected it can be seen that the vector gain margin improves for the closed loop system in $100hz$ and $1000hz$ sampled systems. While in the case of $10hz$ sampled system, the gain margin shrinks due to SOFC while trying to meet response specifications.

The bode plot for the transfer functions y/d and u/d for sampled motor rotor plants are presented in Fig. 15, 16 & 17. From Fig 15 & 16, it can be seen that the transfer function y/d is stable for $100hz$ and $1000hz$ which means the additive disturbance T_d will not make the system unstable. Although the y/d is stable for $100hz$, the gain at the operating range of $0 - 20hz$ is very high and T_d has a significant effect on the steady state performance of the system. However in the case of $10hz$ as y/d is unstable, even a small external disturbance T_d can lead the system to be unstable.

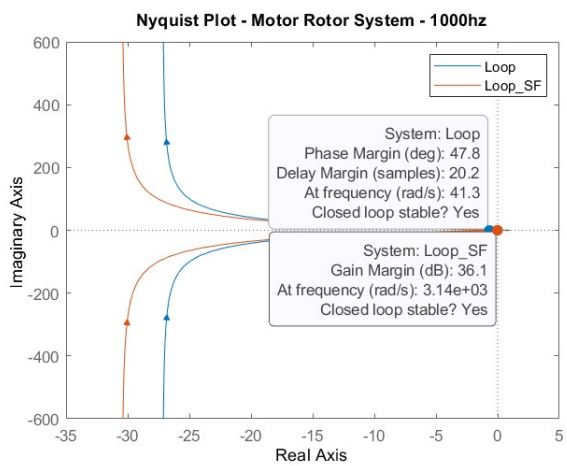

Figure 9: Nyquist Plot for Motor Rotor System Sampled at 1000hz

5 Hardware and Simulation results

The graphs showing the simulation and the hardware implementation for a triangular, square, and sine reference signal are presented in Fig. 18-35.

6 Robustness Analysis

The observer and feedback gains designed for motor rotor model was applied to the stable and unstable pendulum plants to empirically analyze the robustness of the gains. We observed that the gains could track

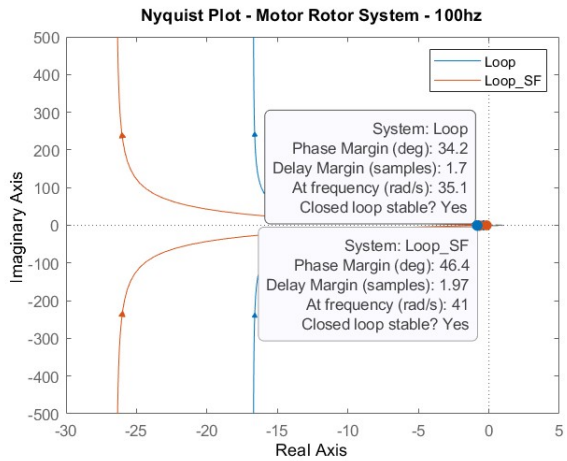

Figure 10: Nyquist Plot for Motor Rotor System Sampled at 100hz

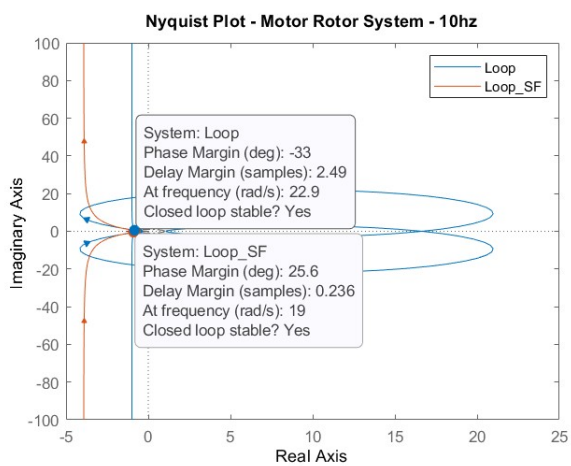

Figure 11: Nyquist Plot for Motor Rotor System Sampled at 10hz

the reference nicely for the stable pendulum case as in Fig. 37 & 39, but not so much for the unstable pendulum case as in Fig. 36 & 38. This might be due to the fact that the open loop inverted pendulum plant by nature is unstable while the gains were designed for a stable system, and even a slight error in the input would lead to large deflections in angle position.

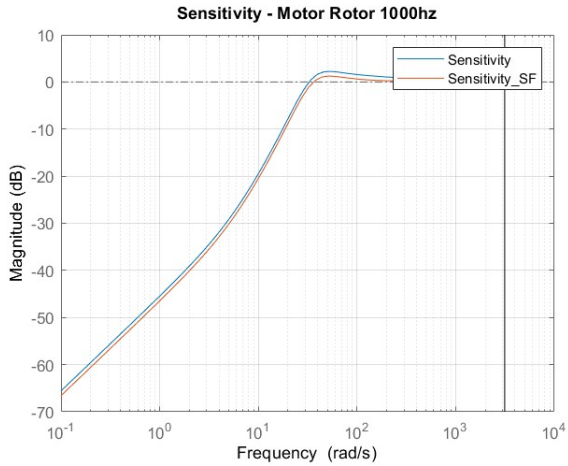

Figure 12: Sensitivity Bode Plot for Motor Rotor System Sampled at 1000hz

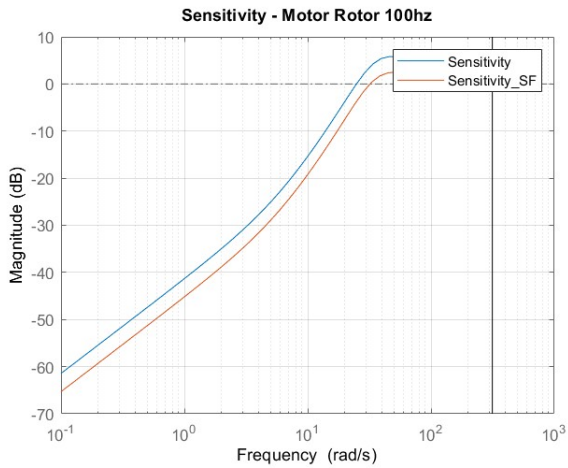

Figure 13: Sensitivity Bode Plot for Motor Rotor System Sampled at 100hz

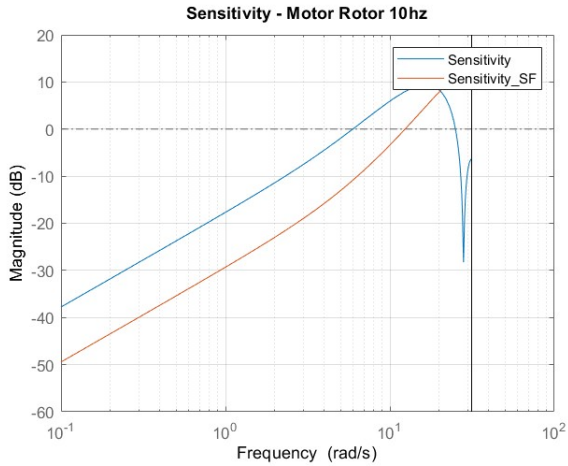

Figure 14: Sensitivity Bode Plot for Motor Rotor System Sampled at 10hz

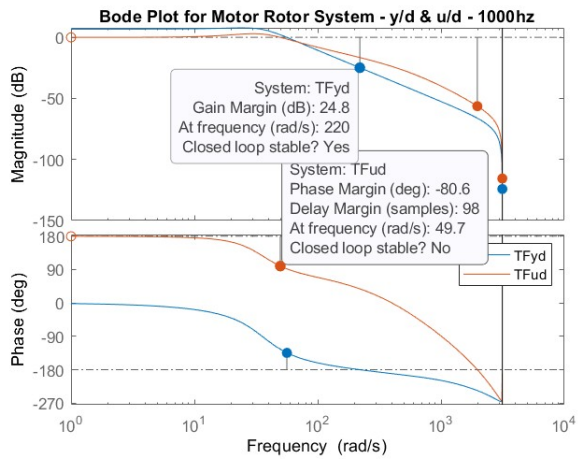

Figure 15: Bode plot of y/d & u/d for Motor Rotor System Sampled at 1000hz

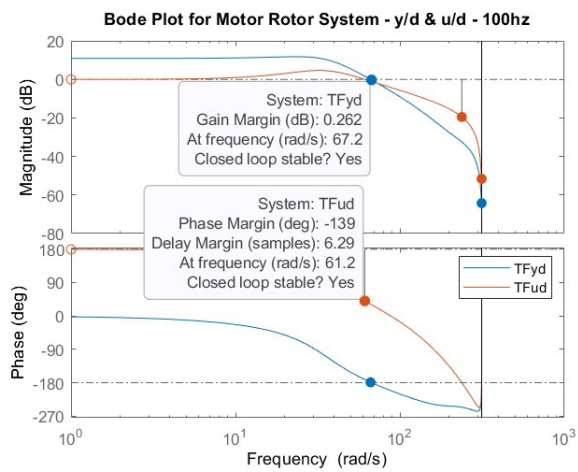

Figure 16: Bode plot of y/d & u/d for Motor Rotor System Sampled at 100hz

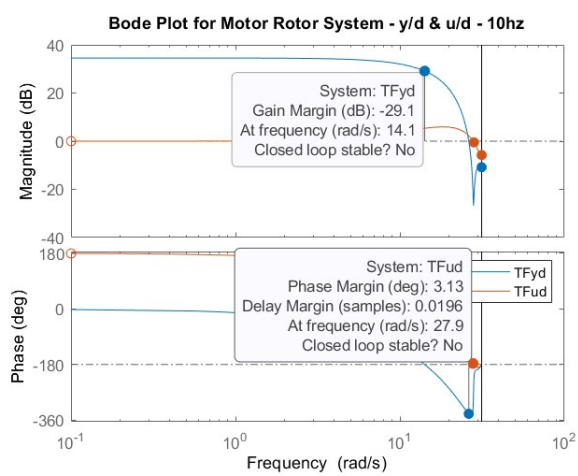

Figure 17: Bode plot of y/d & u/d for Motor Rotor System Sampled at 10hz

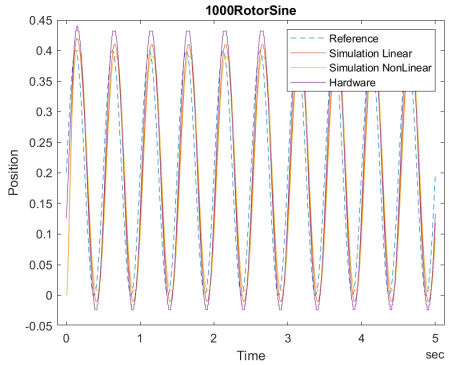

Figure 18: Sine Reference - Rotor - 1000Hz

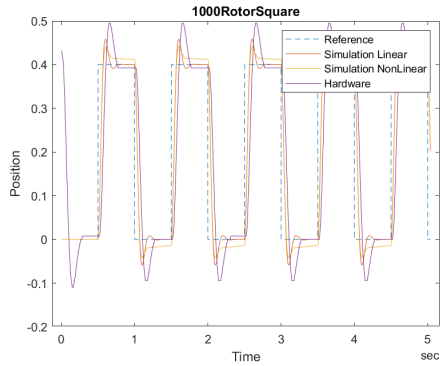

Figure 19: Square Reference - Rotor - 1000Hz

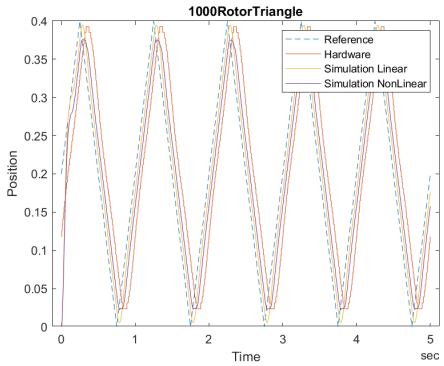

Figure 20: Triangle Reference - Rotor - 1000Hz

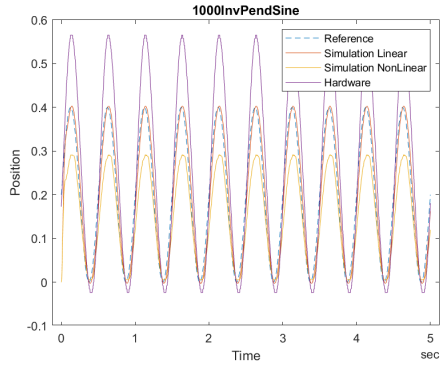

Figure 21: Sine Reference - Inverted Pendulum - 1000Hz

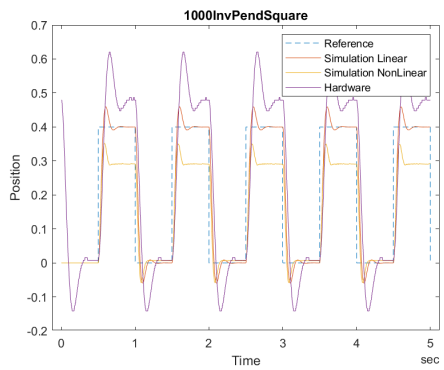

Figure 22: Square Reference - Inverted Pendulum - 1000Hz

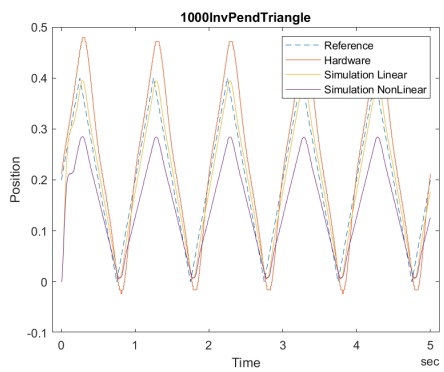

Figure 23: Triangle Reference - Inverted Pendulum - 1000Hz

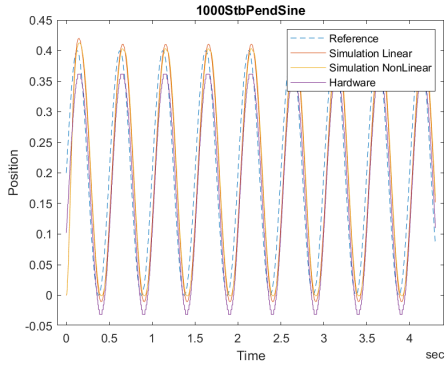

Figure 24: Sine Reference - Stable Pendulum - 1000Hz

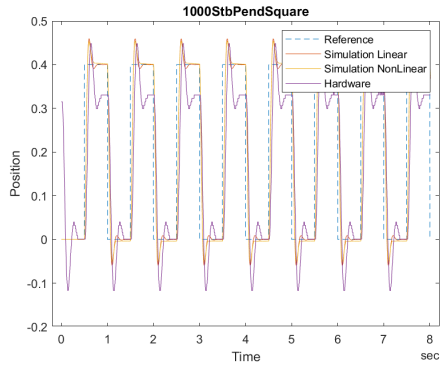

Figure 25: Square Reference - Stable Pendulum - 1000Hz

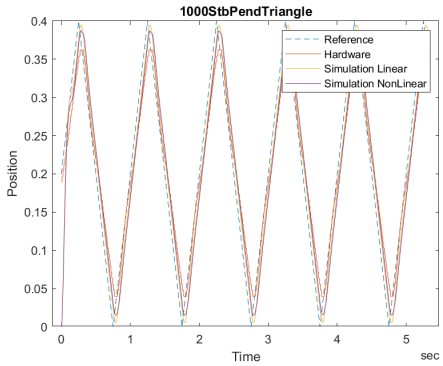

Figure 26: Triangle Reference - Stable Pendulum - 1000Hz

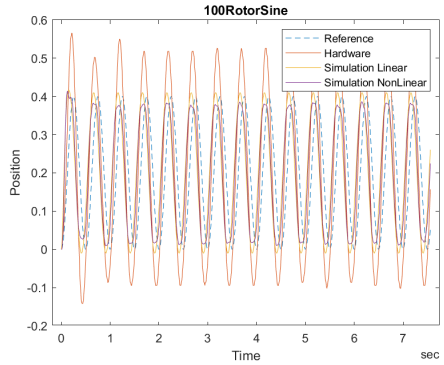

Figure 27: Sine Reference - Rotor - 100Hz

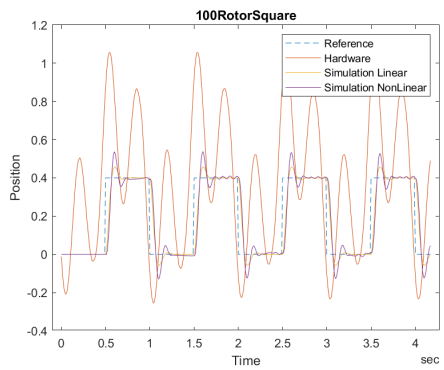

Figure 28: Square Reference - Rotor - 100Hz

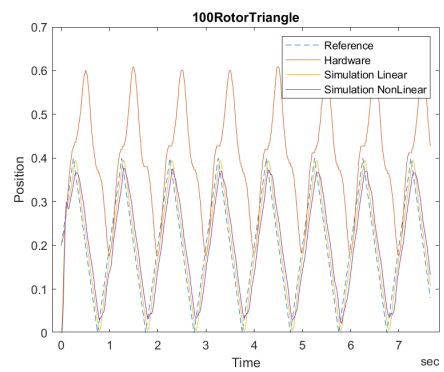

Figure 29: Triangle Reference - Rotor - 100Hz

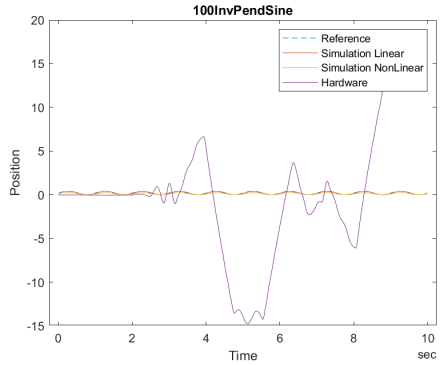

Figure 30: Sine Reference - Inverted Pendulum - 100Hz

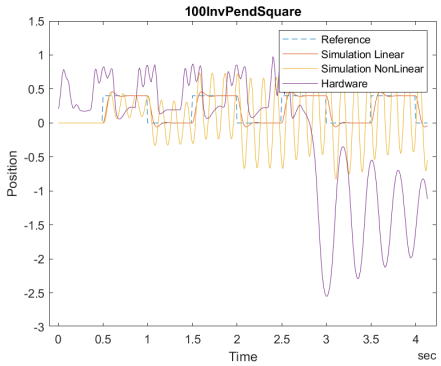

Figure 31: Square Reference - Inverted Pendulum - 100Hz

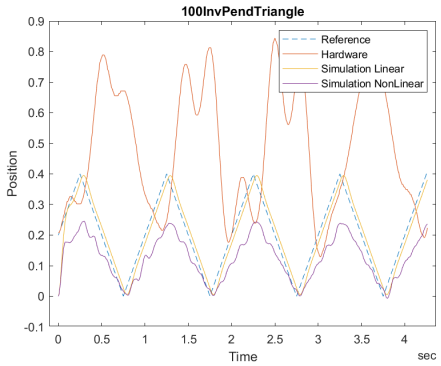

Figure 32: Triangle Reference - Inverted Pendulum - 100Hz

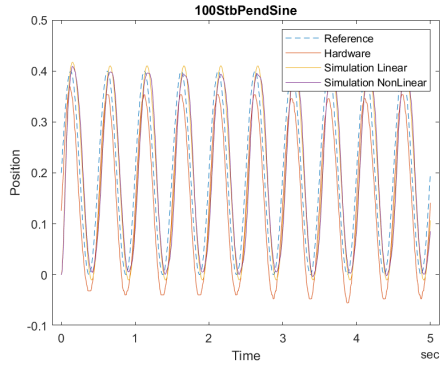

Figure 33: Sine Reference - Stable Pendulum - 100Hz

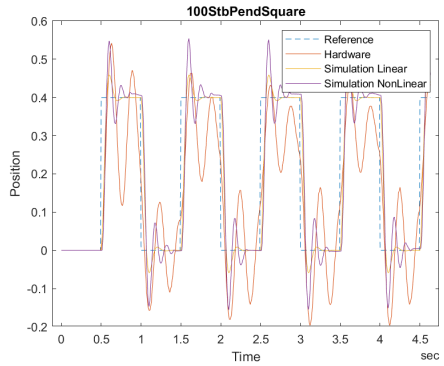

Figure 34: Square Reference - Stable Pendulum - 100Hz

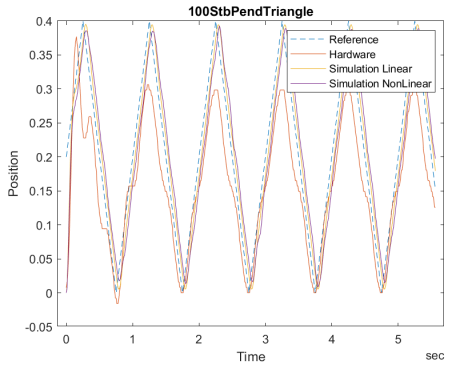

Figure 35: Triangle Reference - Stable Pendulum - 100Hz

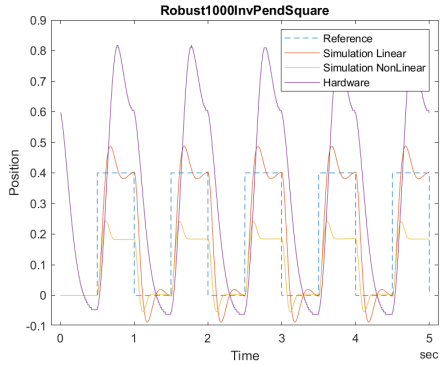

Figure 36: Robustness Analysis - Square Reference - Inverted Pendulum - 1000Hz

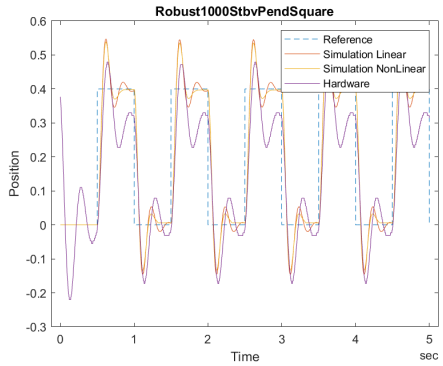

Figure 37: Robustness Analysis - Square Reference - Stable Pendulum - 1000Hz

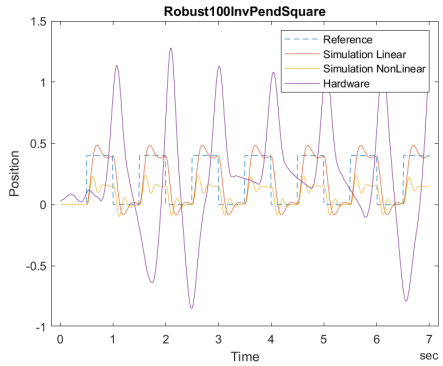

Figure 38: Robustness Analysis - Square Reference - Inverted Pendulum - 100Hz

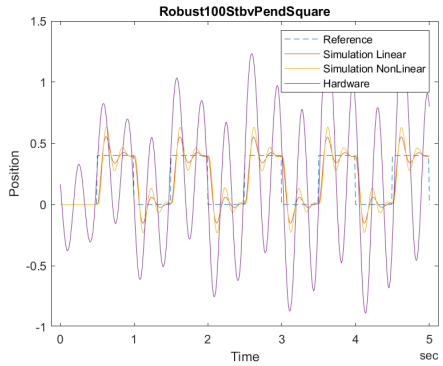

Figure 39: Robustness Analysis - Square Reference - Stable Pendulum - 100Hz